

Space-charge distortion of ionization profile monitor measurements in the CSNS rapid-cycling synchrotron

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Abstract

The ionization profile monitor (IPM) of the CSNS rapid-cycling synchrotron records the transverse proton distribution by collecting residual-gas electrons or ions. During their drift to the collector the secondaries are deflected by the bunch space-charge field, so that the detected r.m.s. width differs from the beam size. Particle tracking with Virtual-IPM 2.3.1 is used to quantify that error for the CSNS geometry and to assess the available mitigations. In electron mode a guiding field of 300 G reduces the relative expansion to the one-percent level for all beam families, intensities, cage voltages and orbit offsets examined; the design field of 0.1 T is conservative. In ion mode the same-sign kick always inflates the profile: at 100 kW a 10 mm beam is expanded by 92% (H_2^+ , injection) and 104% (H_2^+ , extraction). Raising the cage voltage from 5 to 30 kV reduces but does not remove the bias. The true size can be recovered by inverting a species-resolved look-up of the expansion against bunch charge and cage voltage. The recommended operating point is electron collection at $B \gtrsim 300$ G.

Keywords

ionization profile monitor; space charge; CSNS RCS; electron collection; beam-size measurement

1 Introduction

An ionization profile monitor infers the transverse beam density from residual-gas ionization products collected on a position-sensitive detector [1, 4]. The CSNS rapid-cycling synchrotron (RCS) IPM [2, 3, 4, 5] uses a 220×231 mm cage biased at a design voltage of 25 kV ($E_y \simeq 108 \text{ kV m}^{-1}$) and a magnet specified at 0.1 T (available to 0.2 T). Electrons are collected at the top electrode; the three ion species resolved in the time-of-flight spectrum— H_2^+ , H_2O^+ and N_2^+ —are collected at the bottom. Two proton bunches circulate. At 100 kW each carries $N = 7.8 \times 10^{12}$ particles (scaled linearly with beam power P). Injection is at 80 MeV ($\beta = 0.39$, $\sigma_t = 120$ ns, painted $\sigma_0 \simeq 25$ mm or a 10 mm core); extraction is at 1.6 GeV ($\beta = 0.93$, $\sigma_t = 20$ ns, $\sigma_0 \simeq 10$ mm).

While the secondaries drift they experience the bunch electric field and the Lorentz-boosted bunch magnetic field. The detected r.m.s. width σ_m is therefore not the beam width σ_0 . The same mechanism has been analysed for magnetic electron IPMs by Vilsmeier, Sapinski and Storey [7] and for ion IPMs by Shiltsev [8] and by the ISIS group [9]. The question for CSNS is quantitative: how large is the error, how does it grow with bunch charge and shrink with beam size, and how efficiently do the available mitigations—guiding B , cage voltage, and a species-resolved correction—remove it?

The calculation uses Virtual-IPM 2.3.1 [6] with uniform cage fields and 10^5 secondaries per case. Distortion is isolated by comparing bunch space charge on and off at fixed guiding E and B ,

$$\Delta = \frac{\sigma_{\text{SC}} - \sigma_{\text{off}}}{\sigma_{\text{off}}}. \quad (1)$$

A positive Δ is an inflated size; a negative Δ is compression. Microchannel-plate saturation, electromagnetic interference, secondary electrons from the ion trap and the measured cage map are not included. Those hardware effects, rather than space charge, are identified in Refs. [2, 3, 5] as the present electron-mode limit once a sufficient guiding field is applied.

2 Space-charge interaction with collected secondaries

Characteristic times fix the regime. At 25 kV an electron crosses the 115.5 mm half-gap in $\tau \simeq 3.5$ ns. The same path takes $\tau_2 \simeq 210, 631$ and 787 ns for H_2^+ , H_2O^+ and N_2^+ . The time to leave a 10 mm beam is $\tau_0 \simeq 74, 221$ and 275 ns. Injection bunches (120 ns) are longer than τ_0 for H_2^+ ; extraction bunches (20 ns) are shorter than every τ_0 . The bunch spacing is 980 ns at injection and 409 ns at extraction.

Electrons are attracted to the proton bunch. A moderate kick focuses them towards the axis (compression). When the bunch field exceeds the cage field they cross the axis and the collected profile is wider than the beam (over-focusing). Peak transverse bunch fields on a 10 mm beam are approximately 29 kV m^{-1} (injection) and 73 kV m^{-1} (extraction) at 100 kW, versus 145 and 363 kV m^{-1} at 500 kW; the cage field is 108 kV m^{-1} . At 500 kW extraction electrons are trapped in the beam potential during the passage, the regime identified in Ref. [7] as requiring a magnetic field rather than an increase of V .

A guiding field converts the kick into cyclotron motion. If the impulse is delivered early in the flight, the displacement at the collector is proportional to $\sin(\omega_c \tau)/\omega_c$ and vanishes whenever $\omega_c \tau = n\pi$. Successive zeros of $\Delta(B)$ are therefore spaced by

$$\Delta B = \frac{\pi m_e}{e\tau} \propto \sqrt{V}. \quad (2)$$

The envelope of the oscillation falls as B increases. Magnetic mitigation is the decay of that envelope, not the choice of a particular zero. The same cyclotron-phase picture is the “one gyration” tuning rule used at GSI and J-PARC [10, 11].

Ions have the same sign as the beam, so the kick is always outward and Δ is always positive. In the impulsive limit (bunch length $\ll \tau_0$) the transverse impulse scales as N/σ_0^2 and the subsequent drift as $\tau_2 \propto 1/\sqrt{V}$, giving [8]

$$h - 1 \equiv \Delta \propto \frac{N}{V \sigma_0^2}, \quad (3)$$

with a mass dependence $\propto M^{-1/2}$. CSNS occupies a mixed regime: extraction is impulsive, whereas at injection H_2^+ leaves the beam while the 120 ns bunch is still passing. Magnetic confinement is an electron-mode instrument [7, 8] and is not treated further as an ion-mode correction.

3 Profile distortion and growth of the measured size

3.1 Electron collection without a guiding field

At $B = 0$ the collected electron width is not a reliable size. On the 10 mm injection beam at 100 kW, Δ changes from expansion to compression at 13 kV (+9% at 12 kV, ≈ 0 at 13 kV, -8% at 14 kV). At 500 kW the same voltage scan never changes sign: Δ remains positive and grows with V from +29% at 5 kV to +83% at 30 kV (Fig. 1). That is the trapping regime of Section 2: raising V does not un-trap a 500 kW bunch. The same intensity-dependent sign reversal is reported at GSI (electrons shrink without B at moderate intensity) [10] and at J-PARC (a factor-of-two shrink in electron mode) [11].

The map $\sigma_m(\sigma_0)$ is not monotonic at $B = 0$: a small beam may appear larger or smaller than a large one, according to whether the kick focuses or over-focuses. The Voitkiv H_2 double differential cross-section used by Virtual-IPM contributes a σ -independent quadrature smear of 2.9–3.2 mm (≈ 2 –2.5 eV per transverse axis), which is +40–48% at $\sigma_0 = 3$ mm and only +1% at 20 mm. One cyclotron period removes that smear (Section 4.1).

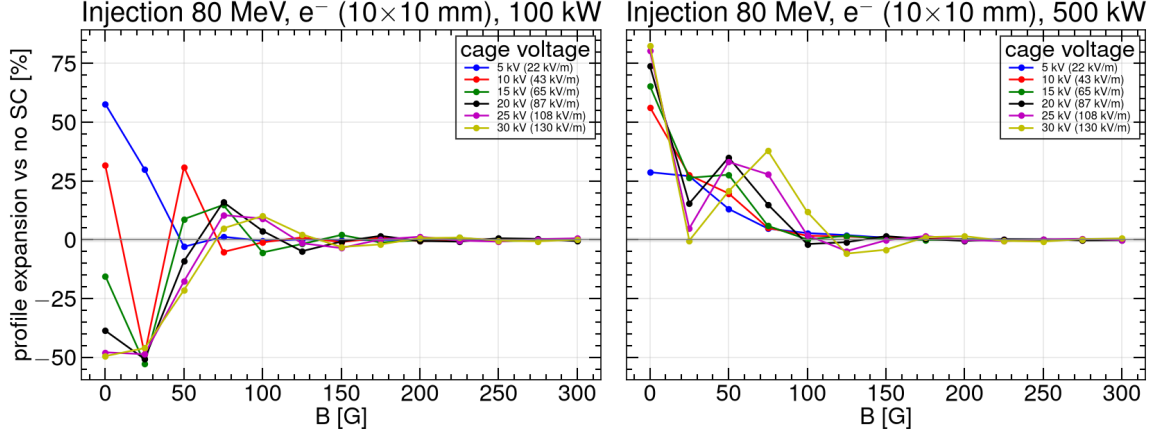


Fig. 1: Relative expansion Δ of the electron profile as a function of guiding field for cage voltages from 5 to 30 kV, on the 10 mm injection beam. At $B = 0$ the 100 kW family crosses zero; the 500 kW family does not.

Along the energy ramp at $B = 0$, a long bunch (120 ns) is compressed and a short bunch (20 ns) is inflated. Both errors become milder as β rises, because the electrons spend less time in the bunch: at 100 kW the 120 ns family goes from -46% (200 MeV) to -32% (1.6 GeV), and the 20 ns family from $+77\%$ (80 MeV) to $+38\%$ (1.2 GeV). Energy alone does not restore the profile.

3.2 Ion collection

Every ion case has $\Delta > 0$. Figure 2 shows σ_m versus σ_0 at the design 0.1 T and 100 kW: every species and both rings lie above the diagonal. Figure 3 shows the growth with bunch charge and the decrease with beam size required by Eq. (3).

For H_2^+ at injection, $\sigma_0 = 10$ mm and 0.1 T, the expansion grows from $+82\%$ (18.2 mm) at 100 kW to $+175\%$, $+279\%$, $+390\%$ and $+422\%$ (52.1 mm) at 200, 300, 400 and 500 kW. Between 100 and 400 kW the growth is $\propto P^{1.12}$, consistent with the linear kick plus the second-order term of $\sigma_m^2 = \sigma_0^2 + \kappa N + \dots$ [8]. At 500 kW the profile reaches the 110 mm half-cage and the exponent saturates.

At extraction, with ions born in time with the bunch, 10 mm, 100 kW and 25 kV, the expansions are $+104\%$ (H_2^+), $+41\%$ (H_2O^+) and $+40\%$ (N_2^+) at $B = 0$, in agreement with a reduced line-charge integration of the same equations to $\pm 1\%$ absolute (Fig. 4; model values $+102\%$, $+40\%$ and $+39\%$). The 20 ns bunch is impulsive, so H_2^+ is the most distorted species. At 500 kW the same family reaches $+408\%$, $+243\%$ and $+235\%$. Between $\sigma_0 = 5$ and 20 mm at 100 kW, $\Delta \propto \sigma_0^{-1.84 \dots -2.01}$, the impulsive σ_0^{-2} of Eq. (3) rather than the continuous-beam $\sigma_0^{-1.5}$. Aligned extraction H_2^+ widths at 0 G and 100 kW are listed in Table 1. A 3 mm extraction beam is already unusable as a size measurement at 100 kW. The most severe case remains injection at 3 mm and 500 kW, where $\sigma_m \approx 56$ mm ($+1700\%$) for all three species.

Table 1: Aligned extraction, H_2^+ , 0 G, 100 kW, 25 kV: obtained width and expansion versus true size.

σ_0 (mm)	3	5	7	10	15	20
σ_m (mm)	38.2	26.8	22.4	20.4	21.7	24.9
Δ (%)	+1172	+435	+219	+104	+44	+24

Species ordering follows the mixed regime. At injection, 10 mm, 0 G and 100 kW the expansions are $+92\%$ (H_2^+), $+65\%$ (H_2O^+) and $+56\%$ (N_2^+): lighter ions expand more, but less than a pure $M^{-1/2}$

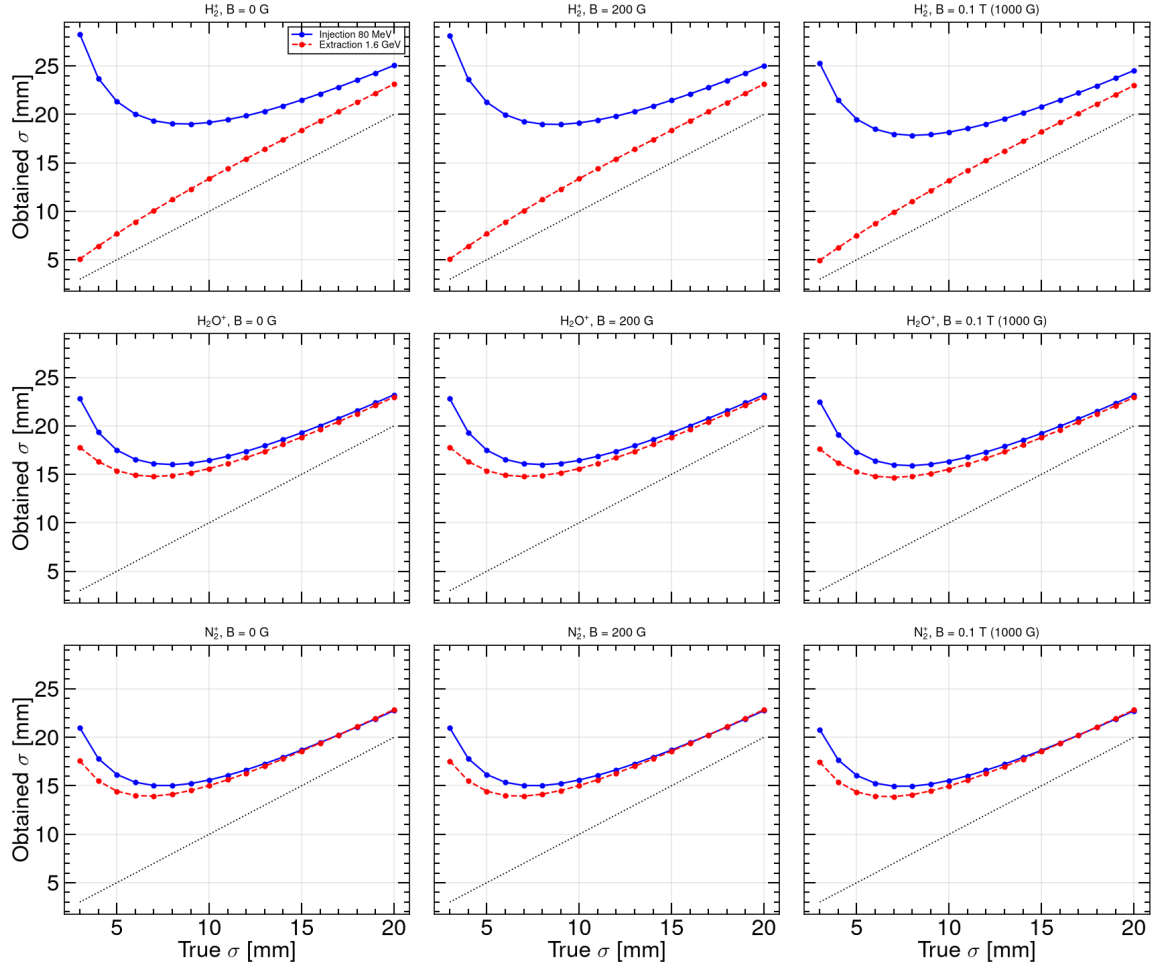


Fig. 2: Ion-mode obtained width σ_m versus true width σ_0 at 0.1 T and 100 kW. The dashed line is $\sigma_m = \sigma_0$. Space charge inflates every point.

law because H_2^+ leaves during the 120 ns bunch. Painted injection (25 mm) is milder (+17%, +10%, +9%). A second extraction bunch, still in the cage when the heavy ions arrive, adds a few points of extra width (single-bunch +36% and +29% versus three-bunch +41% and +40% for H_2O^+ and N_2^+); H_2^+ does not see that bunch (+104% either way). At commissioning power, aligned extraction H_2^+ (10 mm, 0 G) is already +19% at 20 kW and +82% at 80 kW. These magnitudes are of the same order as the operational ion-mode biases reported at J-PARC RCS (20–30% wider than the MWPM) [11] and at the Fermilab Booster [8].

4 Mitigation

4.1 Magnetic confinement of electrons

Figure 5 shows the electron expansion versus B for the three reference beams and 100–500 kW. The curves oscillate through focusing and over-kick. Painted injection recovers first; extraction and the 10 mm injection core still oscillate through 150–250 G. At 300 G every family has settled inside $\pm 1\%$ (Fig. 6).

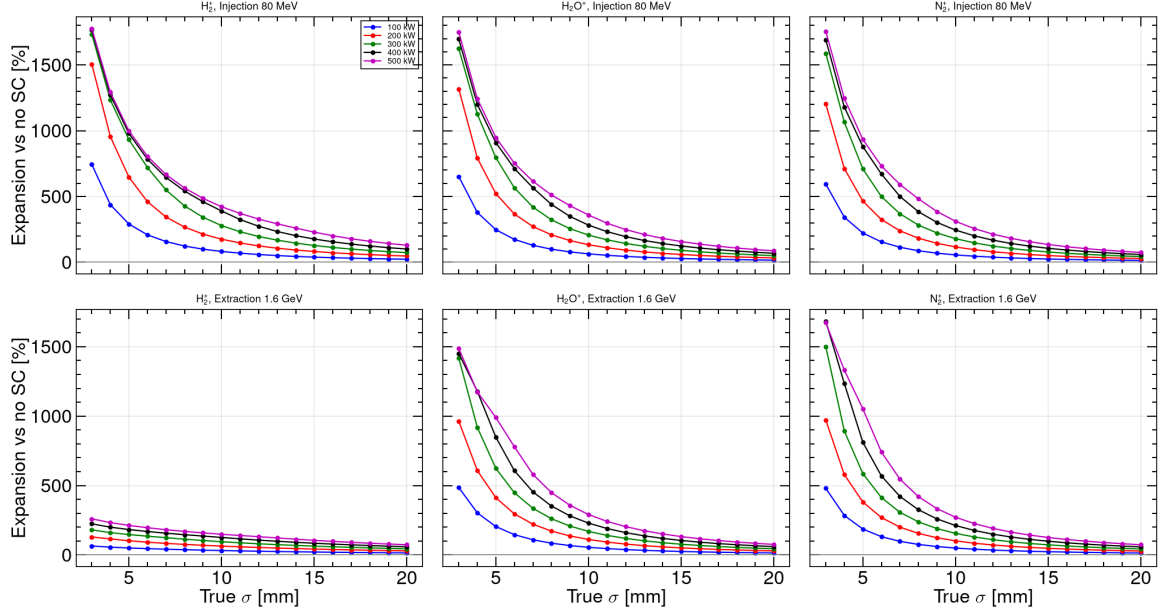


Fig. 3: Ion-mode expansion versus true size at 0.1 T for beam powers from 100 to 500 kW. The 3 mm/500 kW family saturates on the cage wall ($\sigma_m \approx 56$ mm).

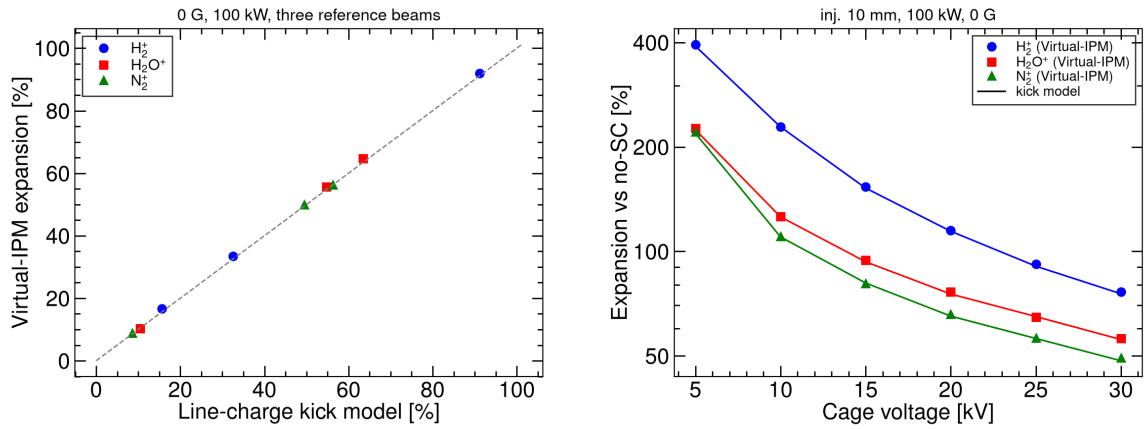


Fig. 4: Left: ion-mode expansion at 0 G and 100 kW, reduced kick model versus Virtual-IPM. Right: cage-voltage scan at 0 G. Agreement is $\pm 1\%$ absolute.

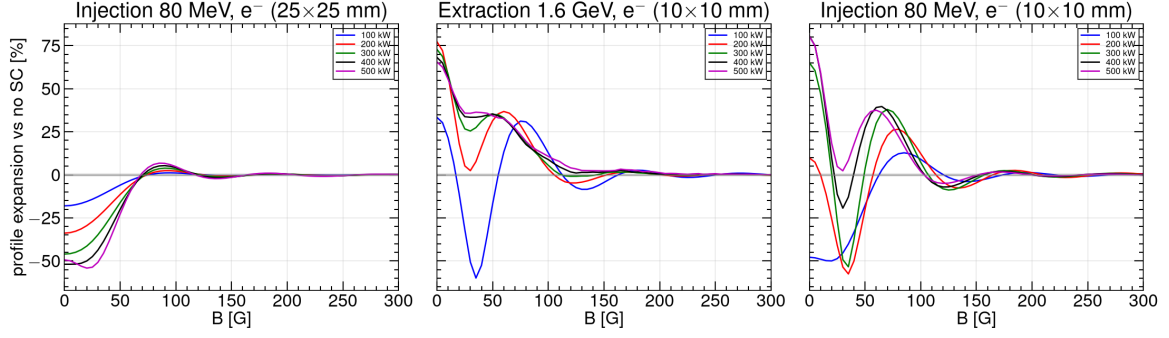


Fig. 5: Electron-mode expansion versus guiding field, 0–300 G, for painted injection (25 mm), extraction (10 mm) and the 10 mm injection core, at 100–500 kW. The grey band is $\pm 1\%$.

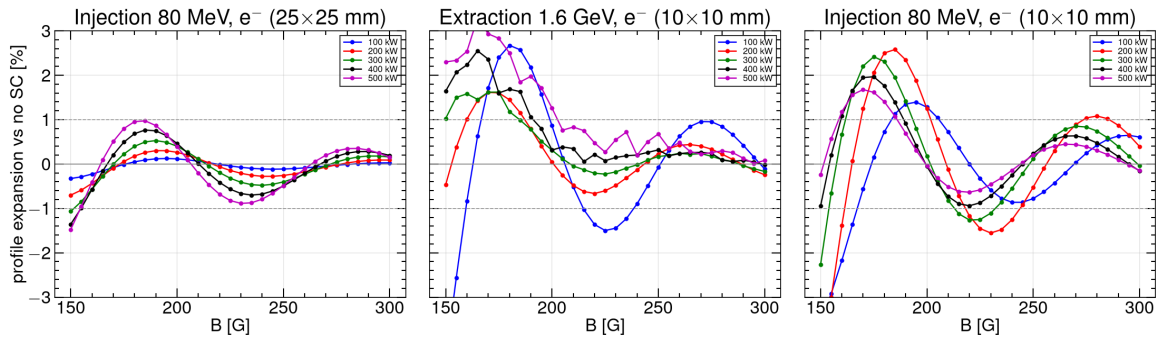


Fig. 6: As Fig. 5, 150–300 G, $\pm 3\%$ vertical scale. At 300 G every family lies inside $\pm 1\%$.

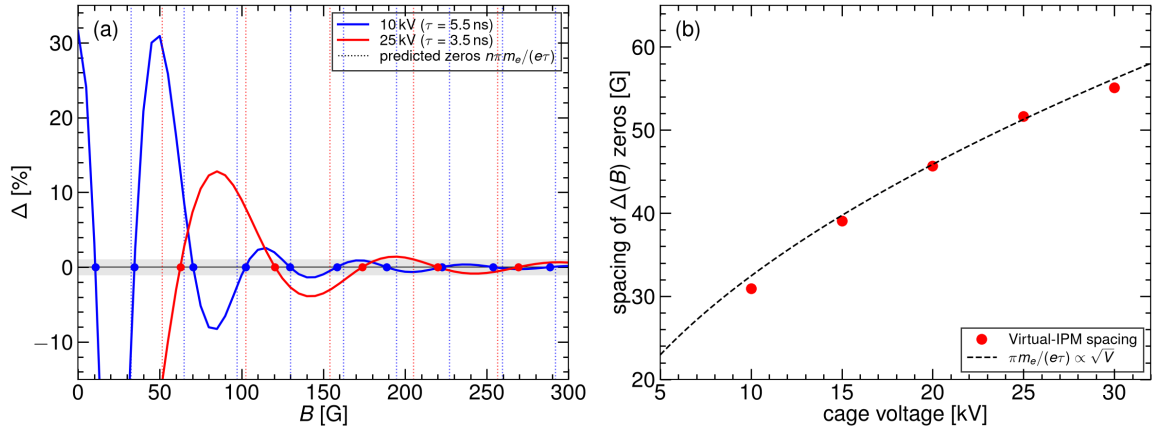


Fig. 7: (a) Relative expansion Δ versus guiding field at two cage voltages (100 kW, 10 mm injection). Dotted vertical lines (colour-matched to the curves) are the predicted zeros $B_n = n\pi m_e/(e\tau)$; circles are the simulated crossings. The grey band is $\pm 1\%$. The 25 kV point at $B = 0$ ($\Delta = -48\%$) is off-scale. (b) Mean spacing of those zeros versus cage voltage, compared with $\pi m_e/(e\tau) \propto \sqrt{V}$ (no free parameter).

Figures 5 and 6 do not fall monotonically: $\Delta(B)$ oscillates through focusing ($\Delta < 0$) and over-kick ($\Delta > 0$). Figure 7 identifies that oscillation as the cyclotron phase of a nearly impulsive space-charge kick.

An electron born on a 10 mm beam leaves the bunch in $\lesssim 1$ ns and then drifts the remaining $\tau \simeq 3.5$ ns (25 kV) in the uniform cage field. The transverse impulse Δv acquired inside the bunch is converted into cyclotron motion at $\omega_c = eB/m_e$. The displacement at the collector is $(\Delta v/\omega_c) \sin(\omega_c \tau)$ and therefore vanishes whenever the electron completes an integer number of half-turns, $\omega_c \tau = n\pi$ ($n = 1, 2, \dots$). Those fields are exactly the equally spaced zeros $B_n = n\pi m_e/(e\tau)$ of Eq. (2). Because $\tau \propto V^{-1/2}$ in a uniform field, the spacing itself scales as $\sqrt{V}$. The amplitude of each successive lobe falls as $1/B$ (the $1/\omega_c$ prefactor). Magnetic mitigation is the decay of that envelope, not the choice of a particular zero: a zero can be moved by a few kilovolts or a change of orbit, whereas the envelope is a property of B .

Panel (a) is a 5 G-step scan at 100 kW and 10 mm injection, drawn at 10 and 25 kV. Vertical dotted lines are the predicted zeros $B_n = n\pi m_e/(e\tau)$ for each voltage; filled circles mark the Virtual-IPM crossings. The 10 kV curve oscillates faster because the flight is longer ($\tau = 5.51$ ns versus 3.48 ns), so a given phase $n\pi$ is reached at a weaker B . The first 25 kV point, $\Delta = -48\%$ at $B = 0$, lies below the frame. At 25 kV the remaining lobes sit near 85, 140, 195, 240 and 295 G with amplitudes 13, 4, 1.4, 0.9 and 0.6%.

Panel (b) extracts the mean spacing between successive zeros at five cage voltages and compares it with $\pi m_e/(e\tau)$ (Table 2). The points follow the $\sqrt{V}$ curve to $\leq 5\%$ from 10 to 30 kV with no free parameter. That is the evidence that the oscillation in Figs. 5 and 6 is the cyclotron phase, and it is why 300 G—the first field at which the envelope itself lies inside $\pm 1\%$ at every power—is the operating point rather than any particular zero.

Table 3 lists the smallest field after which $|\Delta|$ remains below 1%, together with the residual at 300 G. Painted injection is restored first. The last curve to settle is the 10 mm injection core at 200 kW (290 G). A field of 300 G keeps every power at $|\Delta| \lesssim 1\%$.

The same 300 G point holds when the remaining axes are varied. From 200 G the obtained-versus-true plot lies on the diagonal except at 3 mm (Fig. 8). At 300 G, injection at 3 mm is still $+1\%$ (100 kW) to $+4\%$ (500 kW); 0.1 T places 3, 10 and 20 mm on the diagonal at every power ($|\Delta| \leq 0.05\%$). At

Table 2: Electron time of flight from the cage mid-plane and the spacing of $\Delta(B)$ zeros (5 G voltage scan, injection 10 mm, 100 kW). The prediction is Eq. (2) with $\tau = \sqrt{2dm_e/(eE)}$ and $d = 115.5$ mm.

Cage voltage	ToF τ	ΔB predicted	ΔB simulated
10 kV	5.51 ns	32.4 G	30.9 G
15 kV	4.50 ns	39.7 G	39.1 G
20 kV	3.89 ns	45.9 G	45.7 G
25 kV	3.48 ns	51.3 G	51.7 G
30 kV	3.18 ns	56.2 G	55.1 G

Table 3: Smallest guiding field after which $|\Delta| \leq 1\%$ for the remainder of the 0–300 G grid. The residual at 300 G is given in parentheses.

Beam	100 kW	200 kW	300 kW	400 kW	500 kW
Inj. 25 mm	105 G (+0.03%)	115 G (+0.09%)	155 G (+0.16%)	155 G (+0.19%)	155 G (+0.15%)
Ext. 10 mm	240 G (−0.11%)	190 G (−0.24%)	185 G (−0.17%)	195 G (−0.02%)	205 G (+0.08%)
Inj. 10 mm	210 G (+0.61%)	290 G (+0.40%)	235 G (−0.04%)	190 G (−0.16%)	190 G (−0.15%)

300 G and 10–25 kV, both 100 and 500 kW remain $\lesssim 1\%$. Higher V requires slightly more B at 100 kW (150 G at 10 kV to 210 G at 25 kV) because τ is shorter and the first lobes move out; 500 kW recovers earlier at every voltage. Along the energy ramp, 300 G and 100 kW give $|\Delta| \leq 0.5\%$ from 80 MeV to 1.6 GeV for both 120 ns and 20 ns bunches. At commissioning powers (20/50/80 kW) and 300 G the residuals are +0.00/+0.01/+0.02% (painted injection), +0.28/+0.56/+0.18% (extraction 10 mm) and +0.11/+0.29/+0.49% (injection 10 mm).

A few millimetres of vertical orbit offset change Δ by a few percent at $B = 0$ and by $\lesssim 1\%$ at 300 G; the 1% threshold moves by at most ~ 10 G. A horizontal offset is a translation of a uniform cage and does not change the width. The space-charge-induced centroid shift falls as $1/B$ (the $\mathbf{E} \times \mathbf{B}$ / polarisation drift of Ref. [7]): 0.9 mm at 100 G to 0.2 mm at 300 G on the extraction beam.

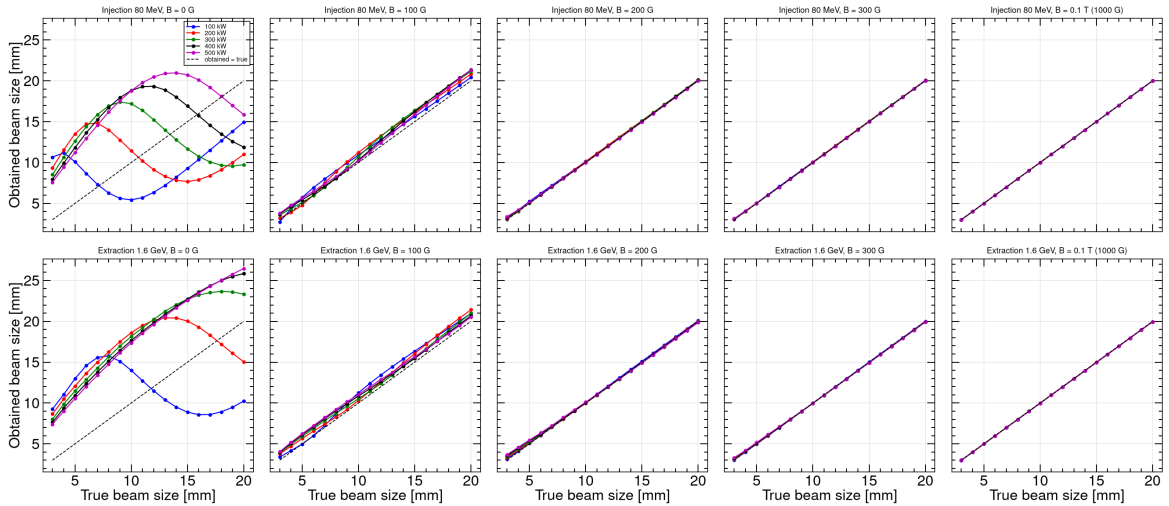


Fig. 8: Electron-mode obtained width versus true width. From 200 G the points lie on the diagonal; 0.1 T includes the 3 mm beams.

Thus $B = 300$ G reduces an electron-mode error of tens of percent (and the wrong sign at 100 kW and high V) to $\lesssim 1\%$. The design 0.1 T provides a factor-of-three margin and is the appropriate field for a 3 mm beam. The CSNS magnet (0.2 T) is not the limiting device. The SNS ring IPM was designed

for 300 G at twenty-five times this bunch charge, with an estimated residual of about 7% [12]; a residual $\lesssim 1\%$ at CSNS is the expected scaling. The CERN PS beam-gas ionization monitor operates at 200 mT for LHC-type bunches [13], at the high-field end of the same scaling.

4.2 Cage voltage

For electrons at $B = 0$, voltage is not a size correction: it can reverse the sign of Δ at 100 kW and cannot reverse it at 500 kW (Section 3.1). Once $B \geq 300$ G the design 25 kV already lies inside 1%, so the voltage may be chosen for collection efficiency and detector operation.

For ions, a higher V shortens τ_2 and reduces the integrated kick, as at ISIS (broadening $\propto 1/E$) [9]. The relevant ion-mode scan is therefore $\Delta(V)$, not $\Delta(B)$: even at the design 0.1 T the ion cyclotron phase is $\omega_c \tau_2 \lesssim 1$ rad, so a guiding field is not a size correction.

Figure 9 is that voltage scan for the three time-of-flight-resolved species on the 10 mm injection beam at 100 kW and $B = 0$. Curves are the line-charge kick model at 1 kV steps from 5 to 30 kV; markers are the Virtual-IPM Block C points on the 5 kV grid. The model tracks every marker to $\leq 4\%$ relative (≤ 4 points absolute). Expansion remains positive at every voltage: from 5 to 30 kV, H_2^+ falls from +395% to +77%, H_2O^+ from +227% to +56%, and N_2^+ from +221% to +49% (Table 4). The Virtual-IPM points scale as $V^{-0.92}$ (H_2^+), $V^{-0.77}$ (H_2O^+) and $V^{-0.83}$ (N_2^+). Aligned extraction H_2^+ (10 mm, 100 kW) falls from +280% at 5 kV to +104% at 25 kV and +93% at 30 kV. Extrapolating $V^{-0.92}$ from the 25 kV injection point, a 10% bias would require approximately 280 kV across the cage, which is not a practical path. ESS can operate an ion IPM at 300 kV m^{-1} because its bunches are weaker by about 5000 [14]; that choice does not transfer to CSNS.

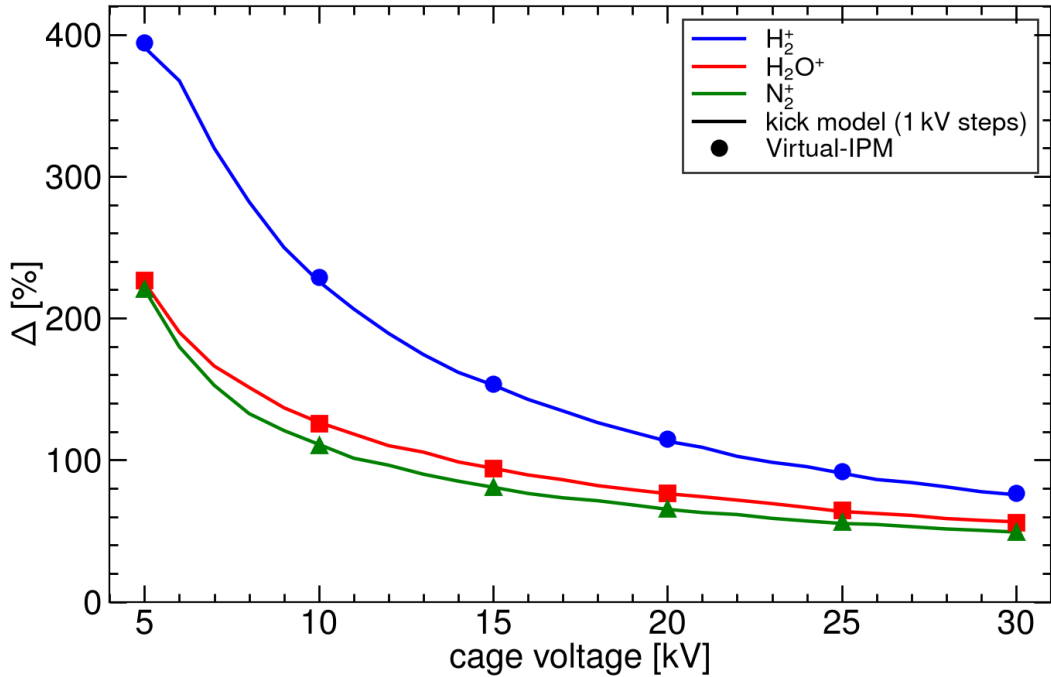


Fig. 9: Ion-mode expansion versus cage voltage for H_2^+ , H_2O^+ and N_2^+ (injection 10 mm, 100 kW, $B = 0$). Curves: kick model at 1 kV steps. Markers: Virtual-IPM. Guiding B is not scanned.

Voltage is therefore a useful lever on the ion bias (a factor of about five for H_2^+ from 5 to 30 kV) but not a substitute for correction or for electron-mode operation.

Table 4: Ion-mode expansion versus cage voltage at $B = 0$ (injection 10 mm, 100 kW). Virtual-IPM Block C.

V (kV)	5	10	15	20	25	30
H_2^+	+395	+229	+153	+115	+92	+77
H_2O^+	+227	+126	+94	+77	+65	+56
N_2^+	+221	+110	+81	+66	+56	+49

4.3 Closed-orbit offset

A few millimetres of closed-orbit error are routine in the RCS. The space-charge kick is centred on the beam, so a horizontal offset Δx is a translation of a uniform cage and must leave the measured width unchanged. A vertical offset Δy changes the drift length $d = 115.5 \text{ mm} + \Delta y$ and therefore the flight time; that is the only first-order geometric lever.

Figure 10 is the electron check on the 10 mm beams at 100 kW. Panel (a) is the Fine D scan $\Delta y = \pm 10 \text{ mm}$ in 1 mm steps. At $B = 0$ extraction expands from +25% ($\Delta y = -10 \text{ mm}$) to +42% ($\Delta y = +10 \text{ mm}$), 0.85% per millimetre, because the electron stays longer in the bunch field when the beam is farther from the collector; injection at $B = 0$ is compressed ($\Delta \simeq -48\%$) and almost flat. At 300 G both beams lie inside the $\pm 1\%$ band over the whole $\pm 10 \text{ mm}$ range ($|\Delta| \leq 0.65\%$). Panel (b) is the horizontal scan at that operating field: $\Delta x = +5$ and $+10 \text{ mm}$ leave Δ unchanged to the line thickness (injection +0.61%, extraction -0.11%). The $\mathbf{E} \times \mathbf{B}$ centroid shift (0.24 mm on extraction at 300 G) is independent of Δx and is a fixed, correctable offset.

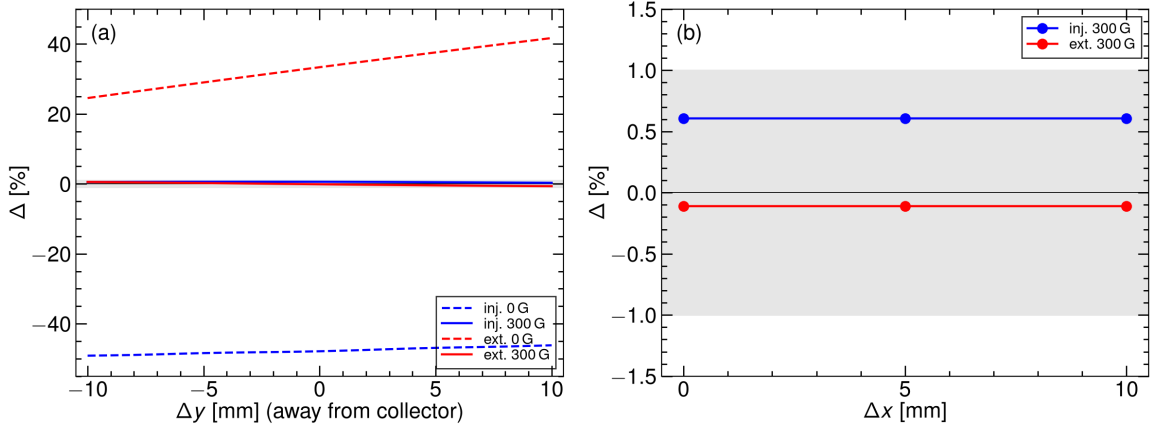


Fig. 10: Electron-mode closed-orbit offset, 10 mm beams, 100 kW. (a) Δ versus Δy (Fine D). Dashed: $B = 0$. Solid: 300 G. The grey band is $\pm 1\%$. (b) Δ versus Δx at 300 G.

Figure 11 is the ion check at $B = 0$ (a guiding field is not an ion-mode lever). Panel (a) plots Δ versus Δy for the three species. Filled markers are injection; open markers are aligned extraction. Lines are $\sigma_m^2 - \sigma_0^2 \propto d$, anchored at the centred Virtual-IPM point, with no free parameter. Injection H_2^+ is +89.2%, +92.0% and +94.7% at $\Delta y = -5, 0$ and $+5 \text{ mm}$; the prediction is +88.9% and +95.0%. Aligned extraction H_2^+ is +101%, +104% and +107% on the same Δy grid. Every species and both beams agree with the drift-length law to 0.5% absolute. Panel (b): $\Delta x = +5$ and $+10 \text{ mm}$ leave every width unchanged to 0.1%, as required by a kick centred on the beam and as stated for ISIS within about 10 mm of the axis [9].

At 500 kW an uncorrected injection H_2^+ profile ($\sigma_m \approx 52 \text{ mm}$) is clipped by the 110 mm half-cage. A 10 mm horizontal offset then pulls the centroid by 4 mm. That is an aperture effect, not a failure of the electron-mode operating point and not a space-charge shift of the ion image.

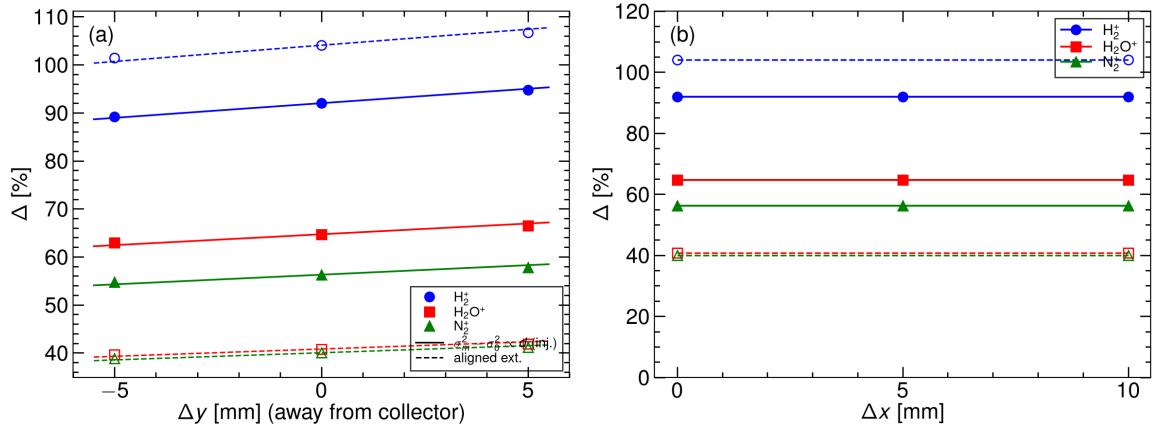


Fig. 11: Ion-mode closed-orbit offset at $B = 0, 10$ mm, 100 kW, 25 kV. (a) Δ versus Δy . Filled: injection. Open: aligned extraction. Lines: $\sigma_m^2 - \sigma_0^2 \propto d$. (b) Δ versus Δx ; the width is translation-invariant. Guiding B is not scanned.

4.4 Inversion of ion-mode widths

The ion bias is large, smooth, and reproduced by the kick model (Fig. 4), so σ_0 can be recovered from σ_m without changing the hardware. On the Fermilab Booster the inversion is $\sigma_m = \sigma_0 h(\sigma_0, N, V)$ with a closed-form h and a single collected species; the iteration is started at $\sigma_0 = \sigma_m$ and is quoted at 5–10% on σ_0 [8]. CSNS cannot copy that recipe. The machine is in the mixed impulsive/continuous regime (Section 2), three ToF peaks are resolved, and $\sigma_m(\sigma_0)$ is not one-to-one. The correction used here is built from the Virtual-IPM Block B / I2 tables at $B = 0, 25$ kV.

Figure 12(a) is the collected width versus the true size for the three species at injection and 100 kW. Each curve has a minimum (H_2^+ : $\sigma_m = 19.01$ mm at $\sigma_0 = 9$ mm; H_2O^+ : 16.01 mm at 8 mm; N_2^+ : 15.02 mm at 8 mm). A single measured σ_m above that minimum has two roots. The closed-form Shiltsev law $\Delta \propto N/(V\sigma_0^2)$ has only the large root of $\sigma_m = \sigma_0 + \kappa/\sigma_0$ and cannot represent the minimum at all.

A concrete case: injection H_2^+ at 100 kW gives $\sigma_m = 19.16$ mm for a true 10 mm beam (dotted line in Fig. 12a). The same collected width is also produced by $\sigma_0 \simeq 8$ mm. Between 7 and 11 mm the H_2^+ width changes by only 0.4 mm, so a single-species inversion is ill-conditioned exactly at the design core. Starting the Booster iteration at $\sigma_0 = \sigma_m$ selects the large root and returns 9.7 mm here only because both roots sit next to the minimum; for a 5 mm beam ($\sigma_m = 21.3$ mm) the same start returns 14.7 mm.

CSNS resolves the three ToF peaks, so two independent widths are available on every shot. The ratio $\sigma_m(H_2^+)/\sigma_m(N_2^+)$ falls monotonically from 1.35 at 3 mm to 1.10 at 20 mm (Fig. 12b). That ratio inverts uniquely. The operational sequence is therefore: identify the H_2^+ and N_2^+ ToF peaks; read σ_m of each and the DCCT bunch charge N ; interpolate $R(\sigma_0, N, V) = \sigma_m^{H_2^+}/\sigma_m^{N_2^+}$ from the Block B / I2 table (or from the kick model); return $\sigma_0 = R^{-1}$. Cage voltage and a BPM Δy enter as parameters of the same table ($\Delta \propto 1/V$ and $\sigma_m^2 - \sigma_0^2 \propto d$, Sections 4.2 and 4.3). The extraction column must be the aligned generation (Fig. 14; H_2^+ +104% at 10 mm, 100 kW), not the v1 as-run +33.5%.

Figure 13 applies that correction to the Virtual-IPM points themselves, leaving the tested σ_0 out of the interpolant. Panel (a), injection 100 kW: the uncorrected σ_m lies far above the diagonal; the single-species large root recovers $\sigma_0 \gtrsim 10$ mm but maps every smaller beam onto the right-hand branch; the H_2^+/N_2^+ ratio recovers 4–19 mm to a median 0.33% (maximum 2.7% at 4 mm). The 10 mm design point returns 10.03 mm. Panel (b): the same two-species invert at 200 and 300 kW stays inside $\pm 2\%$ wherever

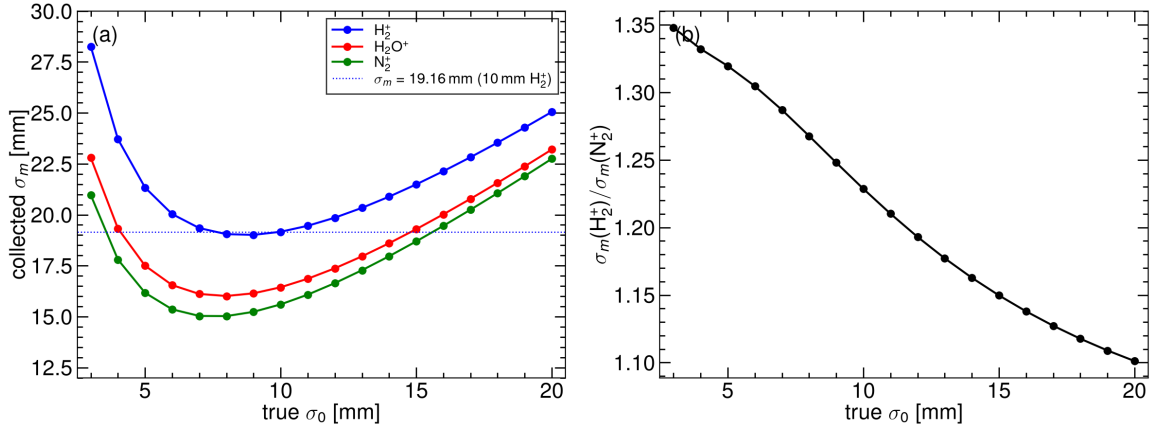


Fig. 12: Injection, 100 kW, 25 kV, $B = 0$. (a) Collected σ_m versus true σ_0 . The dotted line is the H_2^+ width of a 10 mm beam (19.16 mm); it cuts the H_2^+ curve twice. (b) Ratio $\sigma_m(\text{H}_2^+)/\sigma_m(\text{N}_2^+)$, monotonic in σ_0 .

the profile is inside the cage (14 and 10 sizes respectively). At 400 and 500 kW every H_2^+ family with $\sigma_0 \leq 20$ mm is clipped ($\sigma_m \simeq 42\text{--}56$ mm, detected fraction < 0.995) and no table recovers σ_0 .

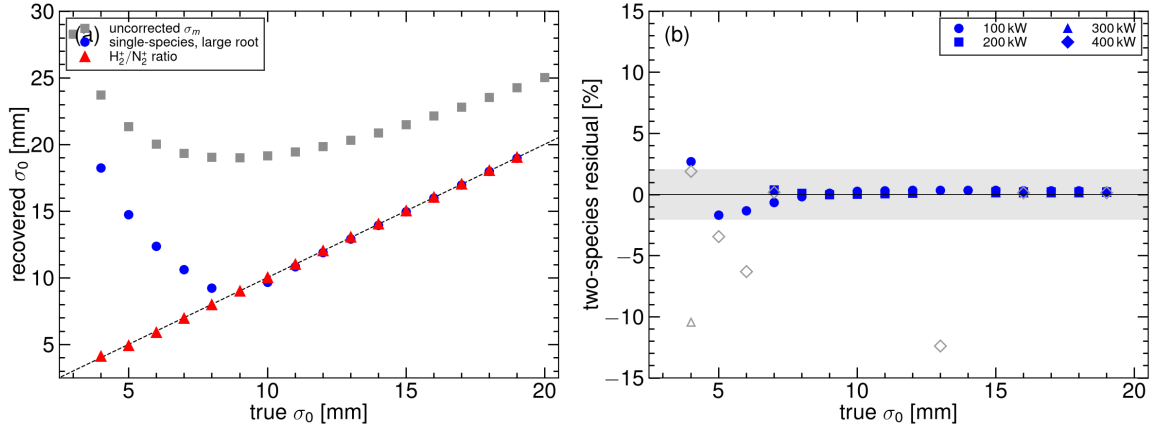


Fig. 13: (a) Injection 100 kW: uncorrected σ_m , single-species large root, and $\text{H}_2^+/\text{N}_2^+$ inversion, each versus true σ_0 (leave-one-out). (b) Two-species residual at 100–300 kW. The grey band is $\pm 2\%$. Open markers are wall-clipped families.

Aligned extraction at 100 kW has the same two-valued $\sigma_m(\sigma_0)$ (H_2^+ minimum 20.4 mm at the 10 mm design size). The I1 size grid is too coarse (3, 5, 7, 10, 15, 20, 25 mm) for a leave-one-out ratio invert at the 0.3% injection level; a production table should be filled at 1 mm in σ_0 , as Block B already is at injection. Field-cage non-uniformity—J-PARC reported a factor-of-two shrink from the external map alone [11]—is not in the table and will have to be folded in once the CSNS measured map is available.

4.5 Summary

Table 5 collects the efficiencies. Magnetic confinement at 300 G is the only mitigation that returns a faithful electron size. Cage voltage reduces the ion bias but does not bring an uncorrected 10 mm, 100 kW profile to the 10% level. The $\text{H}_2^+/\text{N}_2^+$ inversion does, at 100–300 kW, provided the ions stay inside the cage.

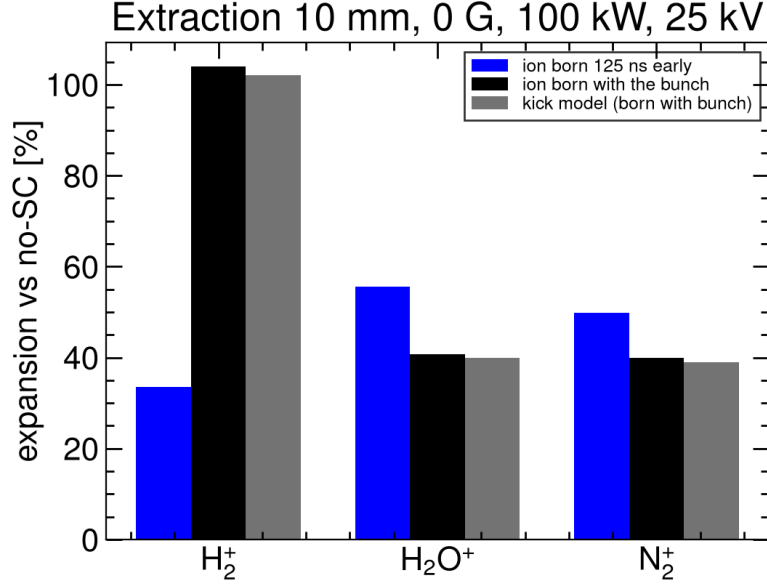


Fig. 14: Extraction, 10 mm, 0 G, 100 kW, 25 kV. Virtual-IPM with the ion born in time with the bunch matches the kick model. This is the extraction column of the inversion table.

Table 5: Efficiency of the mitigations examined. Residuals are quoted for the families of Sections 3 and 4.

Mitigation	Electron mode	Ion mode
$B = 300$ G	tens of percent $\rightarrow \lesssim 1\%$ (all P , V , energy; $\sigma \geq 4$ mm)	—
Design $B = 0.1$ T	3 mm on the diagonal; residual $\leq 0.05\%$	—
Raise V (5 $\rightarrow$ 30 kV)	not a size fix at $B = 0$; unused once $B \geq 300$ G	+395% $\rightarrow$ +77% (H_2^+ , inj. 10 mm, 100 kW, $B = 0$); 10% would require ~ 280 kV
Orbit $\lesssim 10$ mm	300 G still $\lesssim 1\%$	Δx : none; Δy : a few percent
Invert σ_m $\sigma_0 h(\sigma_0, N, V)$	= unnecessary at ≥ 300 G	single species is two-valued; H_2^+/N_2^+ recovers σ_0 to 0.3% (inj. 100 kW, 4–19 mm); fails at the wall ($P \gtrsim 400$ kW)

5 Conclusions

Space-charge distortion of the CSNS RCS IPM has been quantified with uniform-field tracking and checked against a reduced kick model and against the published electron-mode and ion-mode scalings.

Electrons experience an attractive kick that can focus or over-focus. The resulting $\Delta(B)$ is a cyclotron-phase oscillation whose envelope falls below 1% at 300 G for every beam, power, voltage and offset examined, and along the energy ramp. Size growth with charge, and the non-monotonic $\sigma_m(\sigma_0)$ at $B = 0$, both disappear at that field. The design 0.1 T is a margin, except for the smallest (3 mm) beams.

Ions experience a repulsive kick and are always collected with $\sigma_m > \sigma_0$. The inflation scales as N/σ_0^2 (impulsive) and as approximately $1/V$, and matches the kick model to 1%. Raising the cage voltage cannot bring a 10 mm, 100 kW ion profile to the 10% level. A look-up inversion can, but a single ToF peak is two-valued near the design size; the H_2^+/N_2^+ ratio recovers σ_0 to 0.3% at 100 kW provided

the profile has not reached the wall.

The recommended operating point is therefore electron collection at $B \gtrsim 300$ G. Ion time-of-flight should be used for species identification; ion-mode widths should be reported only after inversion. Once space charge is removed, the remaining electron-mode systematic is hardware: microchannel-plate saturation after $\sim 200 \mu\text{s}$, electromagnetic interference, secondary electrons, and the real cage map [2, 3, 5].

Acknowledgements

The hardware parameters follow Ref. [4]. Tracking was performed with Virtual-IPM 2.3.1 [6]. This note is prepared for internal review.

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